NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Arthroplasty

1. THA approaches and complications
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THA approaches and complications

1. Indications and pre-operative planning

- **Indications:** end-stage hip arthritis (primary OA, inflammatory arthritis, AVN, post-traumatic, DDH, Perthes/SCFE sequelae), **displaced femoral neck fractures in active elderly**, failed fixation, tumours.
- **Contraindications:** active infection (joint or systemic), neuropathic (Charcot) joint (relative), severe abductor deficiency (relative: use constrained/dual mobility), medically unfit.
- **Templating** (calibrated AP pelvis, magnification about 20%): **restore the centre of rotation, femoral offset and leg length**; cup at the **teardrop** (medial border) and inferior at the teardrop line; femoral component size and neck cut level; **offset** (reduced offset weakens the abductors and increases impingement/dislocation; excessive offset causes trochanteric pain).
- **Leg length:** measured to the **inter-teardrop line** from the lesser trochanters; **lengthening over about 2-4 cm** risks **sciatic nerve palsy** (peroneal division).

2. Surgical approaches

Approach	Interval / technique	Advantages	Risks
Posterior (Southern/Moore, Kocher-Langenbeck)	Split gluteus maximus; release **short external rotators (piriformis, gemelli, obturator internus) and posterior capsule**	**Commonest**, extensile, preserves abductors, good femoral exposure	**Higher dislocation (posterior)** (reduced by **capsule and rotator repair**, about 1% with repair vs 4% without), **sciatic nerve** injury

Approach	Interval / technique	Advantages	Risks
Direct lateral (Hardinge, transgluteal)	**Split gluteus medius** (anterior third to half) and vastus lateralis (no true internervous plane)	**Low dislocation**, good exposure	**Superior gluteal nerve** injury if the split goes **over about 5 cm above the greater trochanter tip** (abductor weakness, **Trendelenburg gait/limp**), heterotopic ossification
Anterolateral (Watson-Jones)	Between **tensor fasciae latae and gluteus medius** (both superior gluteal nerve: not a true internervous plane)	Low dislocation	Superior gluteal nerve (inferior branch to TFL), abductor damage
Direct anterior (Smith-Petersen/Hueter)	**Superficial: sartorius (femoral) and TFL (superior gluteal)**; **deep: rectus femoris (femoral) and gluteus medius (superior gluteal)**: true internervous plane	Muscle-sparing, **low dislocation**, supine position (fluoroscopy, leg length), possibly faster early recovery	**Lateral femoral cutaneous nerve (meralgia paraesthetica; up to 15-80% transient numbness)**, **femoral fractures** (calcar, greater trochanter) during broaching, wound problems in obese patients, learning curve, **ascending branch of LFCA** (ligate)
Trochanteric osteotomy (Charnley)	Greater trochanter osteotomised	Wide exposure (complex primaries, revision)	**Trochanteric non-union/escape**, wire problems

- **Long-term outcomes are similar** between approaches; choose by surgeon experience and patient factors.
- **Femoral neurovascular structures** lie anterior: anterior retractors placed too medially over the anterior acetabular rim can injure the **femoral nerve/artery**.

3. Acetabular component position

- **Lewinnek "safe zone": inclination (abduction) 40 ±10 degrees**, **anteversion 15 ±10 degrees** (radiographic). Many dislocations occur within this zone, so the **functional (spinopelvic) position** matters: a **stiff lumbar spine** (fusion, degenerative) changes the safe zone (adjust anteversion, consider dual mobility).
- **Combined anteversion (cup + stem) about 25-45 degrees** (Ranawat test/Dorr).
- **Wallace/Hip-spine:** see spinopelvic notes.
- **Steep inclination (over 50 degrees)** = edge loading, increased wear, **squeaking (ceramic-on-ceramic)**, metal ion release (metal-on-metal), dislocation. **Excessive anteversion** = anterior dislocation; **retroversion** = posterior dislocation.
- **Acetabular screws (Wasielewski quadrant system):** a line from the ASIS through the centre of the acetabulum, and a perpendicular: **posterosuperior quadrant = safe** (screws up to about 25 mm+; sciatic nerve and superior gluteal vessels if too long), **posteroinferior = relatively safe** (sciatic, inferior gluteal, internal pudendal), **anterosuperior = "death zone" (external iliac artery and vein)**, **anteroinferior = obturator nerve and vessels**.

4. Fixation

- **Cemented femoral stems: taper-slip (polished, collarless: Exeter, CPT)** subside within the cement mantle creating radial compressive forces (excellent long-term survival), or **composite-beam (shape-closed, matt, collared: Charnley)** bonded to cement. **Third-generation cementing: pulsatile lavage, distal cement restrictor/plug, vacuum mixing (reduces porosity), retrograde filling with a gun, pressurisation, stem**

centralisers, mantle **2 mm or more** (avoid thin mantle/defects: Barrack grading A-D).

- **Uncemented stems:** fixation by **bone ingrowth (porous coating, pore size 50-400 microns)** or **ongrowth (grit-blasted)**; **hydroxyapatite coating** enhances osseointegration; **micromotion under about 40-150 microns** required. **Stress shielding:** proximal bone loss with **stiff, large, extensively coated stems** (Engh); proximal femoral bone loss (Gruen zones 1 and 7).
- **Cups:** mostly uncemented (press-fit, **under-reaming 1-2 mm**, ± screws); cemented polyethylene cups in elderly/poor bone.
- **Age-related trends:** cemented stems reduce **periprosthetic fracture** risk in the elderly (registries: **uncemented stems have higher early periprosthetic fracture rates over about 75 years**).
- **Gruen zones (femur):** 7 zones on AP (1 greater trochanter, 4 tip, 7 calcar), 14 including lateral; **DeLee-Charnley zones (acetabulum)** I-III.

5. Complications

Dislocation

- **About 1-3% after primary THA**; the commonest cause of early revision in some registries.
- **Risk factors: posterior approach without repair**, **component malposition**, **small head size** (**larger heads (32-36 mm) increase jump distance and impingement-free arc**), **reduced offset/soft tissue tension**, abductor deficiency, **neuromuscular disease, cognitive impairment, alcohol**, **spinal fusion/stiffness**, **revision surgery**, female sex, **previous hip surgery**, **fracture as indication**.
- **Early dislocations** (within 3 months) with well-positioned components: **closed reduction** and precautions/brace (about 2/3 do not recur). **Recurrent or malpositioned:** revise: reposition components, **dual-mobility** cup, **larger head**, **constrained liner** (last resort, higher failure), **trochanteric advancement**, abductor reconstruction.
- **Dual-mobility bearings:** small head within a large mobile polyethylene liner inside a polished shell: large effective head, very low dislocation; **intraprosthetic dislocation** is a unique complication.

Periprosthetic fracture (Vancouver classification, femur)

Type	Location	Stem	Treatment
A_G / A_L	Greater / lesser trochanter	Stable	Non-operative unless displaced (A_G: fix if displaced over 2.5 cm/abductor function)
B1	Around/just below the stem	**Stable (well-fixed)**	**ORIF** (locking plate + cables, ± strut allografts)
B2	Around the stem	**Loose**, good bone stock	**Revision with a long stem** (distal fixation, e.g. tapered fluted modular) bypassing the fracture by 2 cortical diameters
B3	Around the stem	**Loose, poor bone stock**	**Revision with proximal femoral replacement** (elderly) or allograft-prosthesis composite (young)
C	Well below the stem tip	Stable	**ORIF** (overlap the stem to avoid stress risers)

- **Distinguishing B1 from B2 is critical**: **a significant proportion of "B1" are actually loose**: test stability intraoperatively; look for radiolucent lines, subsidence, cement mantle fracture.
- **Intraoperative fractures:** commoner with **uncemented press-fit stems**; calcar cracks: **cerclage cable**.
- **Unified Classification System (UCS)** applies to all periprosthetic fractures (see paediatric/trauma notes for type definitions: A apophyseal, B bed of implant (B1 stable, B2 loose, B3 poor bone), C clear of implant, D dividing two implants, E each of two bones, F facing an implant).
- **Acetabular periprosthetic fractures (Paprosky).**

Leg-length discrepancy

- Commonest cause of litigation; lengthening under 1 cm usually tolerated. **True LLD** (bony) vs **apparent/functional** (pelvic obliquity, contractures). **Sciatic palsy risk with lengthening over about 2.5-4 cm.**

Nerve injury (about 0.5-3%)

- **Sciatic nerve (peroneal division)** commonest (posterior approach, lengthening, DDH/revision, retractor placement, cement); **femoral nerve**

(anterior retractors), **superior gluteal nerve** (lateral approach), **lateral femoral cutaneous** (anterior approach), obturator (screws/cement).

Heterotopic ossification (HO)

- **Brooker classification: I** bone islands in soft tissue; **II** spurs with **over 1 cm** between opposing surfaces; **III** under 1 cm gap; **IV** apparent **bony ankylosis**.
- **Risk factors: ankylosing spondylitis, DISH, previous HO**, hypertrophic OA, Paget disease, **male sex**, **lateral/transtrochanteric approaches**, head injury.
- **Prophylaxis** (high-risk patients): **NSAIDs (indomethacin for about 6 weeks)** or **single-dose radiotherapy (about 7-8 Gy) within 24 hours before or 72 hours after surgery**. Excision of mature HO (normal alkaline phosphatase, bone scan quiescent, usually after 6-12 months) with prophylaxis.

Venous thromboembolism

- **DVT without prophylaxis up to about 40-60%**; fatal PE about 0.1-0.5%. **Prophylaxis: LMWH, rivaroxaban/apixaban/dabigatran (DOACs), or aspirin** (**CRISTAL trial** (2022): aspirin had more symptomatic VTE than enoxaparin after arthroplasty; **PEPPER trial**: aspirin, warfarin, rivaroxaban compared), plus **mechanical prophylaxis and early mobilisation**. Extended prophylaxis about **28-35 days after THA** (about 10-14 days after TKA).

Others

- **Infection (about 1%)**, see PJI notes. **Aseptic loosening** and **osteolysis** (wear particles: see bearing notes), **trunnionosis/taper corrosion** (head-neck junction; metal ions; ALTR), **squeaking** (CoC), **iliopsoas impingement** (cup prominent anteriorly; groin pain on resisted hip flexion; **iliopsoas tenotomy or cup revision**), **greater trochanteric pain syndrome**, **vascular injury**, **bone cement implantation syndrome** (hypotension, hypoxia, cardiac arrest during cementing, especially elderly with fractures), **medical complications**.

6. Special situations

- **DDH (Crowe classification by proximal migration of the head): I under 50%**, **II 50-75%**, **III 75-100%**, **IV over 100%** subluxation (relative to femoral head height). **Hartofilakidis** (dysplasia, low dislocation, high dislocation). Place the **cup at the true acetabulum** (anatomical centre), small cups, structural autograft/medialisation, **subtrochanteric shortening osteotomy** (Crowe IV) to prevent sciatic stretch; femur often **narrow and excessively anteverted** (modular or cemented stems to control version).
- **Protrusio acetabuli** (rheumatoid arthritis, Paget, Marfan, osteomalacia): **medial wall bone grafting**, lateralise the centre of rotation, cages/rings in severe cases.
- **Paget disease:** bleeding (consider pre-op bisphosphonates), deformity (osteotomy), hard sclerotic bone, HO.
- **Sickle cell, post-fixation, post-acetabular fracture, ankylosed hip** (abductor function uncertain), **neuromuscular hips**, **young patients** (bearing choice).

Exam pearl: **Posterior approach = commonest, higher dislocation without capsular repair; Hardinge = superior gluteal nerve 5 cm above the trochanter (Trendelenburg); direct anterior = lateral femoral cutaneous nerve and femoral fractures.**

Exam pearl: **Lewinnek: inclination 40 ±10, anteversion 15 ±10. Anterosuperior acetabular quadrant = external iliac vessels (death zone); posterosuperior = safe.** Larger heads and dual mobility reduce dislocation.

Exam pearl: **Vancouver B1 stable = ORIF; B2 loose = long-stem revision; B3 loose + poor bone = proximal femoral replacement/APC; C = ORIF. Brooker HO I-IV; prevent with indomethacin or radiotherapy (7-8 Gy) in high-risk patients.**

TKA balancing and alignment

1. Alignment concepts

- **Mechanical axis of the lower limb:** centre of the femoral head to the centre of the ankle; normally passes **just medial to the centre of the knee** (about 0-3 degrees varus overall).
- **Anatomical vs mechanical femoral axis:** differ by about **5-7 degrees (valgus)**; the **distal femoral cut is made at about 5-7 degrees valgus to the anatomical axis** (perpendicular to the mechanical axis).
- **Joint line:** the native tibial plateau is in **about 3 degrees varus** relative to the tibial mechanical axis; the distal femur in about 3 degrees valgus relative to the femoral mechanical axis; the **joint line is roughly parallel to the floor in stance** (with feet together).
- **Alignment philosophies:**

Philosophy	Principle	Notes
Mechanical alignment (MA)	Femoral and tibial cuts **perpendicular to their mechanical axes** (neutral HKA, joint line horizontal); balance by **soft-tissue releases**	Traditional gold standard; **neutral axis ±3 degrees** associated with better survival (debated with modern implants)
Anatomical alignment (Hungerford)	Joint line at 3 degrees varus, neutral limb	Historical
Kinematic alignment (KA)	**Resurface the pre-arthritic joint** (restore native joint lines and HKA), **minimal releases**	Implants positioned within the patient's native alignment; outlier limb alignments; satisfaction similar or slightly better in some trials

Philosophy	Principle	Notes
Restricted KA / inverse kinematic / functional alignment	KA within safe limits; robotic gap-based adjustments	**CPAK classification** (Coronal Plane Alignment of the Knee: arithmetic HKA and joint line obliquity; 9 phenotypes) describes constitutional alignment

- **Robotic/navigation** TKA reduces alignment outliers; clinical superiority not yet proven.

2. Bone cuts and component rotation

- **Tibial cut:** perpendicular to the tibial mechanical axis in the coronal plane (MA), **posterior slope about 3-7 degrees** (CR knees more; PS knees less, as excess slope can cause instability in flexion/post impingement). Resect about **8-10 mm from the unaffected (higher) side** (measured resection).
- **Distal femoral cut:** about 5-7 degrees valgus to the anatomical axis (intramedullary guide entry point just medial and anterior to the PCL origin); resection equal to the implant thickness (about 9-10 mm).
- **Femoral component rotation (critical for patellar tracking and flexion gap balance):**
  - **Transepicondylar axis (TEA)** (lateral epicondyle to the **medial epicondylar sulcus**: surgical TEA): **most reliable** rotational landmark.
  - **Whiteside line (AP axis):** from the deepest part of the trochlear groove to the centre of the intercondylar notch; femoral component **perpendicular** to it (unreliable in trochlear dysplasia).
  - **Posterior condylar axis (PCA):** femoral component **externally rotated about 3 degrees** relative to it (because the medial posterior condyle is larger); **beware a hypoplastic lateral condyle in valgus knees** (would internally rotate the component: use TEA/Whiteside).
  - **Gap balancing technique:** rotate the femoral component so the **flexion gap is rectangular** after the tibial cut with balanced ligaments.
- **Tibial component rotation:** aligned to the **medial third of the tibial tubercle** (Akagi line: PCL attachment to the medial border of the patellar tendon at its tibial attachment).

- **Internal rotation of femoral or tibial components** (combined) causes **patellar maltracking, lateral subluxation, anterior knee pain, patellar complications and stiffness** (Berger: combined internal rotation 1-4 degrees = tilt; 3-8 = subluxation; 7-17 = dislocation/failure). Assess with **CT**.
- **Anterior femoral notching** (anterior cortex): increases **supracondylar periprosthetic fracture** risk (about 30-40% reduction in torsional strength).
- **Femoral sizing: anterior referencing** (cuts the posterior condyles by the measured amount; downsizing changes the flexion gap: may loosen flexion) vs **posterior referencing** (restores posterior condylar offset; risk of anterior notching or overstuffing the patellofemoral joint).

3. Gap balancing

- **Extension gap** (distal femoral cut + tibial cut) and **flexion gap** (posterior femoral cut + tibial cut) must be **equal and rectangular**.
- **The tibial cut affects both gaps; the distal femoral cut affects only the extension gap; the posterior femoral cuts/femoral size affect only the flexion gap.**

Problem	Flexion gap	Extension gap	Solution
Tight in both	Tight	Tight	**Cut more tibia**
Loose in both	Loose	Loose	**Thicker polyethylene insert** (or augments)
Tight in extension, OK in flexion	Balanced	Tight	**Release posterior capsule / remove posterior osteophytes**, then **resect more distal femur** (about 2 mm)
Loose in extension, OK in flexion	Balanced	Loose	**Distal femoral augmentation** (add distal femur), thicker insert + smaller/posterior cut (combined)
Tight in flexion, OK in extension	Tight	Balanced	**Downsize the femoral component** (anterior referencing) or **increase tibial slope**; release PCL (CR)

Problem	Flexion gap	Extension gap	Solution
Loose in flexion, OK in extension	Loose	Balanced	**Upsize femur** (posterior augments) or **thicker insert + more distal femoral resection**

- **Joint line:** keep within about **8 mm** of native: **elevation of the joint line** (excess distal femoral resection + thick insert) causes **mid-flexion instability, patella baja (pseudo), patellofemoral problems**.

4. Coronal deformity correction (soft-tissue releases)

Varus knee (commonest)

- **Tight medial structures:** remove **medial osteophytes first** (tibial and femoral), then release **deep MCL (meniscotibial)**, **posteromedial capsule/semimembranosus** (tight in **extension**), **superficial MCL** (subperiosteal from tibia or **pie-crusting**; the **anterior sMCL is tight in flexion**, posterior in extension), **pes anserinus** (rarely). **PCL** release (in CR knees with tight flexion).
- Also: **medial tibial reduction osteotomy** (downsize/lateralise the tibial tray and remove medial bone).
- **Fixed flexion deformity:** remove posterior osteophytes, **posterior capsular release**, gastrocnemius recession; as a last resort extra distal femoral resection.

Valgus knee

- **Tight lateral structures: iliotibial band** (tight in **extension**), **popliteus** (tight in **flexion**), **lateral collateral ligament** (tight in **flexion and extension**), **posterolateral capsule** (extension), lateral head of gastrocnemius, biceps femoris.
- **Order (Ranawat inside-out pie-crusting, Whiteside):** posterolateral capsule, ITB, then popliteus/LCL as needed.
- **Hypoplastic lateral femoral condyle**: do not use the PCA alone for rotation.
- **Peroneal nerve palsy** risk when correcting **severe valgus with flexion contracture** (stretch): **flex the knee post-operatively**, loosen dressings;

explore if no recovery.

- **Ranawat valgus types:** I (correctable, medial soft tissue intact), II (fixed, medial attenuated: may need constraint), III (severe osseous deformity with incompetent medial structures: constrained implant).

5. Constraint and design choices

Design	Mechanism	Indications
Cruciate-retaining (CR)	PCL retained; flatter or ultra-congruent inserts	Intact, functional PCL; less bone resection; **paradoxical anterior femoral sliding**
Posterior-stabilised (PS)	**Cam-post** substitutes the PCL (**femoral rollback**)	**Absent/incompetent PCL**, **patellectomy**, inflammatory arthritis, large deformity, previous HTO (debated); risks: **post wear/fracture, patellar clunk syndrome** (fibrous nodule at the superior pole catching the box), cam jump (flexion instability)
Medial pivot / ultra-congruent	Highly conforming medial compartment	Mimics native kinematics
Varus-valgus constrained (CCK, non-linked)	Tall, wide post in a deep box	**MCL/LCL incompetence**, severe valgus, flexion-extension mismatch, revisions
Rotating-hinge (linked)	Hinged femur-tibia linkage with rotation	**Global ligamentous deficiency**, severe bone loss, distal femoral replacement (tumour/fracture), hyperextension instability, **neuromuscular recurvatum** (polio)

- **Use the least constraint needed** (more constraint = more stress transfer to fixation interfaces = loosening).
- **Mobile-bearing vs fixed-bearing**: no proven difference in survival; spinout risk with mobile bearings.
- **Patellar resurfacing:** reduces **reoperation for anterior knee pain** (meta-analyses, KAT trial: little functional difference); inflammatory arthritis

favours resurfacing. **Patellar thickness should be restored** (avoid overstuffing). **Patellar complications:** maltracking (component internal rotation, lateral placement of femoral/tibial components, medial placement of patellar button is preferred), **fracture** (over-resection, avascular after lateral release), **avascular necrosis**, **patellar clunk**, loosening, tendon rupture.

- **Patellar tracking: "no thumb" test**; lateral retinacular release if needed (protect the **superior lateral genicular artery**).

6. Complications of TKA (non-infective)

- **Stiffness/arthrofibrosis** (flexion under about 90 degrees at 6-12 weeks): **manipulation under anaesthesia (within about 6-12 weeks)**, arthrolysis, revision.
- **Instability: flexion instability** (flexion gap larger than extension; recurrent effusions, PS cam jump), **mid-flexion instability**, **genu recurvatum**, global instability: revise with balanced gaps/increased constraint.
- **Aseptic loosening/osteolysis**, **polyethylene wear**.
- **Periprosthetic fractures: supracondylar femur (Lewis-Rorabeck I undisplaced, II displaced + stable prosthesis (ORIF: locking plate or retrograde nail if open box), III loose prosthesis (revision/DFR))**; **tibial (Felix)**; **patella (Goldberg/Ortiguera)**.
- **Peroneal nerve palsy** (valgus, flexion contracture, epidural anaesthesia), **vascular injury** (popliteal artery behind the posterior capsule: **posterolateral corner, about 9 mm from the posterior tibia in flexion**).
- **Wound complications: use the most lateral previous incision** (lateral skin flap less perfused: blood supply from the medial side), skin bridges at least about 5-6 cm.
- **Extensor mechanism disruption:** patellar tendon rupture (**worst; augment with allograft or synthetic mesh, medial gastrocnemius flap**), quadriceps rupture.
- **VTE**, **persistent pain** (about 10-20% dissatisfied after TKA: highest predictors are pre-operative expectations, pain catastrophising, mild radiographic OA).

- **Unexplained painful TKA** work-up: **exclude infection first (ESR/CRP, aspiration)**, then instability, malrotation (CT), loosening, patellar issues, referred pain (hip, spine), CRPS, neuroma.

Exam pearl: **Tibial cut affects both gaps; distal femoral cut affects extension only; posterior femoral cut/femoral size affects flexion only. Tight both = cut more tibia; loose both = thicker insert.**

Exam pearl: **Femoral rotation: TEA most reliable; 3 degrees external to the posterior condylar axis; Whiteside line perpendicular. Internal rotation of components = patellar maltracking. Tibial rotation: medial third of the tubercle.**

Exam pearl: **Varus: medial osteophytes, deep MCL, posteromedial capsule/semimembranosus, then superficial MCL. Valgus: ITB (extension), popliteus (flexion), LCL (both); peroneal palsy risk with valgus + flexion contracture.** Least constraint necessary; rotating hinge for global ligament loss.

Bearing surfaces and wear

1. Tribology basics

- **Wear:** removal of material from articulating surfaces. **Mechanisms:**
  - **Adhesive** (commonest in metal-on-polyethylene: polymer transfers to the counterface and is pulled off),
  - **Abrasive** (hard asperities or particles plough the softer surface),
  - **Third-body** (trapped particles: cement, bone, metal debris),
  - **Fatigue/delamination** (subsurface cracking; important in **TKA**, low-conformity, oxidised polyethylene),
  - **Corrosive** (fretting at modular junctions).
- **Modes of wear (McKellop): Mode 1** intended bearing surfaces; **Mode 2** a primary surface against a non-intended secondary surface (e.g. head against shell after liner wear-through); **Mode 3** third-body particles between primary surfaces; **Mode 4** two non-intended surfaces rubbing (back-side wear of a liner against the shell, neck against the cup rim, **trunnion corrosion**).
- **Lubrication: fluid-film** (ideal; surfaces fully separated: ceramic-on-ceramic and well-designed metal-on-metal with large heads and low clearance), **boundary** (surfaces in contact, lubricant adsorbed: metal-on-polyethylene), **mixed**. Lambda ratio (fluid-film thickness / combined surface roughness) **over 3 = full fluid-film**.
- **Linear wear** (penetration of the head into the liner, mm/year) vs **volumetric wear** (mm^3/year: **determines particle load and osteolysis**). **Volumetric wear increases with head size** ($V = \pi r^2 w$, Charnley); **linear wear decreases with larger heads**.
- **Osteolysis threshold** for conventional polyethylene: **about 0.1 mm/year linear wear** (Dumbleton); above this, osteolysis is common.

2. Biology of particle disease (osteolysis)

- **Wear particles (0.1-1 micron polyethylene most bioactive)** are phagocytosed by **macrophages**, which release **TNF-alpha, IL-1, IL-6, PGE2**, upregulate **RANKL** on osteoblasts/stromal cells: **osteoclast activation** and

bone resorption. The **effective joint space** (Schmalzried) lets particles migrate along the implant-bone interface.

- **Clinical:** often **silent** until large lesions, loosening or fracture; **serial radiographs**, **CT** for extent (pelvic lesions underestimated on radiographs).
- **Treatment: liner exchange and bone grafting** of lytic lesions with retained well-fixed components (risk of dislocation after isolated liner exchange: use larger heads), or full revision for loose components.

3. Polyethylene

- **UHMWPE (ultra-high molecular weight polyethylene):** semi-crystalline polymer. **Gamma irradiation** (sterilisation) breaks chains forming **free radicals**: in air, these **oxidise** (embrittlement, **subsurface white band**, delamination, worse with **shelf ageing**).
- **Sterilisation:** gamma in **inert gas/vacuum** packaging, **ethylene oxide** or **gas plasma** (no cross-linking, no free radicals).
- **Highly cross-linked polyethylene (HXLPE):** irradiated at **5-10 Mrad** to cross-link chains (**much lower adhesive/abrasive wear**: **wear reduced about 80-90%**, dramatic fall in osteolysis and revision for wear in THA); **post-irradiation thermal treatment** removes free radicals: **remelting (above the melting point, about 137 C: eliminates free radicals but reduces crystallinity, fatigue strength and fracture toughness)** or **annealing (below melting: preserves mechanical properties but leaves some free radicals: in-vivo oxidation)**. **Vitamin E (alpha-tocopherol)-infused/blended HXLPE** quenches free radicals without remelting.
- **Cross-linking reduces fatigue strength and toughness**: concern in **TKA (high contact stress, fatigue wear)** and thin liners, **rim/locking mechanism fractures** with thin liners and large heads or malposition.
- **Polyethylene thickness: at least 6-8 mm** in TKA (thinner = high contact stress, delamination).
- **Compression-moulded (direct)** polyethylene has better wear performance than **ram-extruded + machined** stock in TKA; **Hylamer** (failed); **calcium stearate**-free resins.
- **TKA wear:** fatigue-dominated (**delamination, pitting**) because of **low conformity**, rolling/sliding, high contact stresses; **backside wear** of

modular inserts (micromotion against the tray) contributes to tibial osteolysis.

4. Metal-on-polyethylene (MoP)

- **Cobalt-chromium or ceramic heads on HXLPE**: the commonest and most proven bearing in THA (excellent long-term registry survival).
- **Head size: 32-36 mm** heads with HXLPE balance dislocation risk and wear; very thin liners for large heads risk fracture.

5. Ceramics

- **Alumina (Al_2O_3)** and **zirconia-toughened alumina (BIOLOX delta, ZTA)**: **hardest, most scratch-resistant, very low wear, inert particles, excellent wettability (hydrophilic: fluid-film lubrication)**, **lowest wear of all bearings (ceramic-on-ceramic)**.
- **Problems: brittleness** (fracture: much rarer with delta ceramics, about 0.002-0.02%), **squeaking** (about 1-3%: associated with **cup malposition (edge loading)**, **stripe wear**, impingement), **chipping on insertion** (malseating), **limited neck lengths**, cost.
- **Ceramic head fracture:** revise to a **ceramic head** (not metal: **third-body ceramic particles** cause catastrophic metal head wear) with full synovectomy; use a new taper sleeve if the trunnion is damaged.
- **Ceramic-on-polyethylene:** reduces polyethylene wear vs metal heads (smoother, harder, more wettable), avoids taper corrosion (trunnionosis) of CoCr heads.
- **Oxidised zirconium (Oxinium):** zirconium alloy with a **ceramic oxide surface layer**: low wear against polyethylene with metal toughness; **avoid against a damaged surface**: if the thin layer is breached (e.g. after dislocation and closed reduction over the cup rim), the soft substrate wears rapidly.
- **Old zirconia (yttria-stabilised) heads** suffered **phase transformation** (Prozyr recall 2001).

6. Metal-on-metal (MoM)

- **Cobalt-chromium on cobalt-chromium**; large heads possible (hip resurfacing: **Birmingham Hip Resurfacing**; large-head THA). Low volumetric wear in ideal conditions but **nanometre-sized metal particles** (huge numbers) and **metal ions (cobalt, chromium)**.
- **Adverse reactions to metal debris (ARMD) / adverse local tissue reaction (ALTR): pseudotumours** (solid or cystic masses), **ALVAL** (aseptic lymphocyte-dominated vasculitis-associated lesion: type IV delayed hypersensitivity), **metallosis**, **soft-tissue necrosis** (abductor destruction). **Systemic cobalt toxicity:** cardiomyopathy, neuropathy, hypothyroidism, visual/hearing loss.
- **Risk factors: female sex**, **small component size**, **steep cup inclination (over about 50-55 degrees: edge loading)**, **specific designs (ASR recall 2010)**, large-head MoM THA (trunnion corrosion), bilateral implants, renal impairment.
- **Surveillance (MHRA/FDA): blood cobalt and chromium levels** (concern **over about 7 µg/L (ppb)**; some use 2-4 µg/L for unilateral resurfacing as threshold for closer monitoring), **MARS MRI** (metal artefact reduction sequence) or ultrasound for pseudotumours.
- **Revision** for symptomatic patients, rising ions, enlarging pseudotumours or osteolysis; revise to a **ceramic or polyethylene** bearing. **Outcomes of revision for ALTR are worse** than for other aseptic causes.
- **Hip resurfacing: young, active men with large femoral heads** (best results; **avoid in women, small heads, osteoporosis, large cysts, AVN, DDH**). **Femoral neck fracture** (commonest early failure; **notching** of the neck, varus positioning).
- **Metal hypersensitivity** in TKA (nickel) is debated (rarely a true cause of failure; diagnosis of exclusion).

7. Taper corrosion (trunnionosis)

- **Mechanically assisted crevice corrosion** at the **head-neck (and modular neck-stem) junction**: micromotion disrupts the passive oxide layer: fretting + crevice corrosion: **cobalt release**, **ALTR even with metal-on-polyethylene**.

- **Risk factors: large CoCr heads** (larger moment arm/torque), **long necks/high offset**, **dissimilar metals (CoCr head on titanium stem)**, small/short trunnions, **dual-taper modular necks** (recalled designs), poor taper assembly (blood/fluid on the taper, inadequate impaction).
- **Diagnosis: raised cobalt >> chromium** (Co:Cr ratio over about 3-5 suggests taper corrosion rather than bearing wear), **MARS MRI**.
- **Treatment: revise the head to a ceramic head with a titanium sleeve**, change liner, debride; revise the stem if the trunnion is badly damaged.

8. Choosing a bearing (summary)

Bearing	Wear	Particles	Main concerns	Typical use
MoP (CoCr on HXLPE)	Low (with HXLPE)	Polyethylene (osteolysis)	Taper corrosion with large CoCr heads	**Most patients**
Ceramic-on-HXLPE	Lower	Polyethylene	Cost; ceramic fracture (rare)	Younger/active patients; avoid taper issues
Ceramic-on-ceramic	**Lowest**	Ceramic (inert)	**Squeaking, fracture, chipping**, cup position sensitive	Young active patients
MoM	Low (volumetric) but ions	**Metal nanoparticles/ions**	**ALTR/pseudotumour, systemic cobalt**	Largely abandoned except selected resurfacing

- **Hardness (decreasing):** ceramic > CoCr > stainless steel > titanium > polyethylene. **Titanium is never used as a bearing surface** (poor wear resistance, notch sensitivity).
- **Wettability (hydrophilicity):** ceramics best.

Exam pearl: **Volumetric wear determines osteolysis; larger heads = more volumetric, less linear wear (with conventional PE). Osteolysis threshold about 0.1 mm/year. Particles of 0.1-1 micron activate macrophages, TNF-alpha, RANKL.**

Exam pearl: **HXLPE reduces THA wear about 80-90%; remelting removes free radicals but lowers fatigue strength; annealing keeps strength but leaves free radicals; vitamin E is the alternative.** TKA wear is fatigue/delamination (conformity and thickness at least 6-8 mm).

Exam pearl: **Ceramic-on-ceramic: lowest wear; squeaking and fracture; revise a broken ceramic head with a new ceramic head. MoM: pseudotumour/ALVAL, women, small sizes, steep cups; check cobalt/chromium and MARS MRI. Trunnionosis: cobalt much higher than chromium.**

Periprosthetic infection (ICM criteria)

1. Epidemiology and pathogenesis

- **Periprosthetic joint infection (PJI)** occurs in about **1-2% of primary hip and knee arthroplasties** (higher in revisions, about 3-5%). Among the commonest causes of revision (leading cause of early TKA revision).
- **Biofilm:** bacteria adhere to the implant, produce an **extracellular polymeric matrix (glycocalyx)**; sessile bacteria are **up to 1,000 times more resistant to antibiotics** and protected from host defences. **Biofilm matures within about 3-4 weeks**: the basis for implant retention only in early infections.
- **"Race for the surface"** (Gristina): host cells vs bacteria competing to colonise the implant.
- **Organisms: coagulase-negative staphylococci (S. epidermidis) and S. aureus** (together about 50-60%), streptococci, enterococci, gram-negatives, anaerobes; **Cutibacterium acnes** (shoulder, low-grade), fungi (Candida); **polymicrobial** (about 10-15%); **culture-negative** (about 7-15%: prior antibiotics, fastidious organisms).
- **Routes:** perioperative contamination (most), **haematogenous** (late acute; dental, urinary, skin, endocarditis), contiguous spread.

2. Risk factors

- **Patient: diabetes (HbA1c over about 7.5-8%, perioperative glucose over about 200 mg/dl)**, **obesity (BMI over 40)**, **smoking**, **malnutrition (albumin under 3.5 g/dl, transferrin under 200, lymphocytes under 1,500)**, **rheumatoid arthritis/immunosuppression**, **S. aureus colonisation (nasal carriers)**, chronic kidney/liver disease, **prior joint surgery/infection**, **intra-articular steroid injection within about 3 months**, IV drug use, anaemia, male sex, depression.
- **Surgical:** prolonged operative time, **allogeneic transfusion**, haematoma, persistent wound drainage (over about 5-7 days), revision surgery, concurrent infection elsewhere.

3. Prevention

- **Optimise** glucose, BMI, nutrition, smoking cessation (4-6 weeks), treat dental/urinary infection.
- **MRSA/MSSA screening and decolonisation** (**nasal mupirocin + chlorhexidine body washes**).
- **Prophylactic antibiotics: cefazolin within 60 minutes before incision** (vancomycin within 120 minutes if MRSA-colonised or beta-lactam allergic; clindamycin alternative); **redose** after about 4 hours or with large blood loss (over 1,500 ml); **stop within 24 hours**. Antibiotic-loaded cement in high-risk patients.
- **Skin preparation: chlorhexidine-alcohol** (superior to povidone-iodine in surgical site infection trials), hair clipping (not shaving) just before surgery.
- **Theatre: laminar flow**, restricted traffic, **dilute povidone-iodine lavage** before closure (debated benefit), meticulous haemostasis, tranexamic acid (fewer transfusions).

4. Classification by timing (management-oriented)

Type	Timing	Features	Typical organism
Early post-operative (acute)	**Within about 4 weeks (up to 3 months in some systems)** of surgery	Wound drainage, erythema, fever, pain	S. aureus, gram-negatives
Acute haematogenous	Any time, **symptoms under about 3 weeks** in a previously well-functioning joint	Sudden pain, effusion, sepsis	S. aureus, streptococci
Chronic (late)	**Over about 4 weeks**/insidious	Persistent pain, early loosening, sinus tract	CoNS, C. acnes
Positive intraoperative cultures	Unexpected positive cultures at revision for "aseptic" failure	**Two or more positive**: treat as PJI (about 6 weeks)	CoNS, C. acnes

- **Tsukayama classification:** I positive intraoperative cultures, II early post-operative, III acute haematogenous, IV late chronic.

5. Diagnosis

Clinical

- **Pain** (commonest symptom), stiffness, effusion, wound problems, **sinus tract** (pathognomonic), fever (often absent in chronic infection), **early loosening** (within 5 years).

2018 ICM (International Consensus Meeting / Parvizi) definition

- **Major criteria (either = infected):**
  - **Two positive cultures of the same organism** (by standard culture methods).
  - **A sinus tract communicating with the joint or visualisation of the prosthesis.**
- **Minor criteria (pre-operative scoring):**

Criterion	Score
Serum CRP over 1 mg/dL (10 mg/L) or **D-dimer over 860 ng/mL**	**2**
Serum ESR over 30 mm/h	**1**
Synovial WBC over 3,000 cells/µL or **positive leukocyte esterase (++)**	**3**
Positive alpha-defensin (signal-to-cutoff ratio over 1)	**3**
Synovial PMN over 80%	**2**
Synovial CRP over 6.9 mg/L	**1**

  - **Total 6 or more = infected; 2-5 = possibly infected** (add intraoperative findings); **0-1 = not infected**.
- **Intraoperative findings** (for inconclusive pre-operative scores): **positive histology (over 5 neutrophils per high-power field in 5 or more fields: 3 points)**, **purulence (3 points)**, **single positive culture (2 points)**; combined score **6 or more = infected**, 4-5 inconclusive, 3 or less not infected.

- **Thresholds differ for the early post-operative period** (within 6 weeks): **CRP over 100 mg/L (knee/hip acute)**, synovial WBC over about **10,000**, PMN over about **90%** (inflammation from surgery raises markers).
- **Inflammatory arthritis, crystal disease, metallosis (ALTR)** can give false-positive synovial counts (manual count in metallosis).
- **EBJIS 2021 definition** (European Bone and Joint Infection Society) uses "infection likely/confirmed" tiers with similar markers.

Practical work-up (AAOS/ICM)

1. **Serum ESR and CRP** in every painful arthroplasty (high sensitivity: if both normal, PJI unlikely).
2. If raised (or high suspicion): **joint aspiration** (stop antibiotics at least **2 weeks** before aspiration) for **synovial WBC, PMN%, culture** (aerobic + anaerobic; **hold cultures 10-14 days** if C. acnes or low-virulence organisms suspected), **leukocyte esterase**, **alpha-defensin**.
3. **Intraoperative: 5 (3-6) separate tissue samples** with separate instruments, **frozen section**, **sonication** of explants (improves culture yield, especially after antibiotics), avoid swabs.
4. **Imaging:** radiographs (early loosening, periosteal reaction, osteolysis); nuclear imaging (labelled leukocyte with marrow scan) when aspiration is inconclusive; MRI with MARS sequences.

6. Treatment

Debridement, antibiotics and implant retention (DAIR)

- **Indications: early post-operative (within about 4 weeks)** or **acute haematogenous (symptoms under 3 weeks)** infection, **well-fixed implants**, **healthy soft tissue (no sinus)**, susceptible organism.
- **Technique:** radical **synovectomy and debridement**, high-volume lavage (dilute povidone-iodine, chlorhexidine), **exchange of all modular parts** (head and liner/polyethylene insert: essential), followed by **pathogen-specific antibiotics** (**rifampicin** added for staphylococci with retained implants: active against biofilm; never as monotherapy; **fluoroquinolones for gram-negatives**) for about **3 months (hip) to 6 months (knee)**.

- **Success about 50-80%**; worse with **S. aureus/MRSA**, delays, high CRP, rheumatoid arthritis, prior DAIR. **Repeated DAIR** has poor results. **KLIC and CRIME80 scores** predict DAIR failure.

One-stage exchange

- Remove all components, radical debridement, re-implant new components (often **antibiotic-loaded cement**), specific antibiotics. Increasingly used for **chronic infection** with **known organism and susceptibilities**, healthy soft tissues, no sinus (debated), immunocompetent host (Frankfurt/ENDO-Klinik practice). Similar control rates to two-stage in selected cases, fewer operations.

Two-stage exchange (traditional gold standard for chronic PJI, especially in North America)

1. **Stage 1:** remove all components and cement, radical debridement, **antibiotic-loaded cement spacer** (static or **articulating**; high-dose antibiotics: e.g. **vancomycin + gentamicin/tobramycin**; heat-stable antibiotics, about **3-4 g of antibiotic per 40 g cement**), followed by **4-6 weeks of IV/oral antibiotics**.
2. **Antibiotic holiday** (about 2 weeks; debated) and assessment of infection control (**ESR/CRP trends, aspiration**: no single test reliably confirms eradication).
3. **Stage 2:** re-implantation (often after **6-12 weeks**), intraoperative cultures/frozen section.
4. **Success about 85-90%**. **Interim mortality and failure to reach stage 2** are significant in elderly patients.
5. **After re-implantation:** a further **3 months of oral antibiotics** reduced failure in one RCT (Frank et al.).

Salvage

- **Resection arthroplasty (Girdlestone for the hip)**, **knee arthrodesis** (intramedullary nail or external fixation), **amputation** (uncontrolled infection, life-threatening sepsis, massive bone/soft-tissue loss), **long-term suppressive antibiotics** (unfit patients or retained infected implants).

Antibiotic principles

- Bactericidal, biofilm-active agents with good bone penetration: **rifampicin (staphylococci)**, **fluoroquinolones (gram-negatives)**; adjust to cultures; **monitor toxicity** (rifampicin interactions, vancomycin nephrotoxicity, linezolid myelosuppression).
- **Oral vs IV: OVIVA trial** (2019): **oral antibiotics were non-inferior to IV** within the first 6 weeks for bone and joint infection (including PJI), with fewer line complications.
- **Duration: DATIPO trial** (2021): **12 weeks of antibiotics** was superior to 6 weeks for PJI (DAIR/one-stage/two-stage) in persistent infection rates.

7. Special situations

- **Acute haematogenous infection with a well-fixed implant:** DAIR quickly (ideally within days of symptom onset).
- **Fungal PJI:** two-stage, antifungal-loaded spacer (amphotericin/voriconazole), prolonged antifungals; poor outcomes.
- **C. acnes** (shoulder): slow growing (hold cultures 14 days), male, young, previous shoulder surgery.
- **Antibiotic prophylaxis for dental procedures** in arthroplasty patients: **not routinely recommended** (AAOS/ADA), except selected immunocompromised patients.

Exam pearl: **2018 ICM major criteria: two positive cultures of the same organism, or a sinus tract. Minor: CRP/D-dimer (2), ESR (1), synovial WBC over 3,000 or LE ++ (3), alpha-defensin (3), PMN over 80% (2), synovial CRP (1); 6 or more = infected.**

Exam pearl: **DAIR (with modular exchange) for early post-operative (within 4 weeks) or acute haematogenous infection (under 3 weeks symptoms) with stable implants; add rifampicin for staphylococci. Chronic infection: one- or two-stage exchange.** OVIVA: oral non-inferior; DATIPO: 12 weeks beats 6.

Exam pearl: **Biofilm matures in about 3-4 weeks; CoNS and S. aureus are commonest; stop antibiotics 2 weeks before aspiration; take 5 tissue samples; sonicate implants.** Cefazolin within 60 minutes of incision; chlorhexidine-alcohol skin prep.

Revision arthroplasty

1. Why arthroplasties fail

- **THA (registries): aseptic loosening/osteolysis** (commonest long-term), **dislocation/instability**, **infection**, **periprosthetic fracture**, adverse reaction to metal debris, leg-length/pain.
- **TKA: infection** (commonest cause of early revision), **aseptic loosening** (commonest long-term), **instability**, **arthrofibrosis**, polyethylene wear (historical), periprosthetic fracture, malalignment/malrotation, extensor mechanism failure, unexplained pain.
- **Time-dependence: early (under 2 years)**: infection, instability, malposition, fracture; **late**: wear, osteolysis, loosening.
- **Golden rule: every painful or failing arthroplasty is infected until proved otherwise**: **ESR/CRP ± aspiration** before any revision.

2. Evaluation before revision

- History (pain pattern: start-up thigh pain suggests femoral stem loosening; groin/buttock pain suggests acetabular loosening), wound history, systemic symptoms.
- **Infection work-up** (ICM criteria).
- **Serial radiographs: radiolucent lines over 2 mm, progressive**, **subsidence**, **component migration**, cement fracture, **pedestal formation** (distal endosteal bone around an uncemented stem tip: loose), **spot welds** (bone ingrowth: stable), osteolysis, polyethylene wear (eccentric head position).
  - **Engh criteria** for uncemented stem fixation (bony ingrowth vs fibrous stable vs unstable).
  - **Harris criteria** for cemented femoral loosening: **definite** (migration of the stem, cement fracture, stem fracture), **probable** (continuous radiolucent line around the entire cement mantle), **possible** (lucency over 50-100% of the interface).
- **CT** (osteolysis, component version, bone stock), **MARS MRI** (ALTR), **bone scan** (loosening is non-specific), **metal ions** (MoM, taper).

- **Implant identification**, operative notes, extraction tools, components and allograft availability, **neurovascular** proximity (intrapelvic cement/components: CT angiography).

3. Bone loss classifications

Acetabulum: Paprosky

Type	Rim	Columns	Migration / landmarks	Reconstruction
I	Intact, minimal bone loss	Intact	No migration	**Hemispherical uncemented cup**
II (A, B, C)	Distorted (IIA superomedial, **IIB superolateral (rim deficient under 1/3)**, IIC medial wall)	**Intact (supportive)**	**Superior migration under 3 cm**; teardrop and ischium intact	**Hemispherical (jumbo) cup ± screws, particulate graft**
IIIA ("up and out")	Superolateral rim deficient (30-60% of circumference)	Partially supportive	**Over 3 cm superior migration**, **Kohler line intact**, moderate ischial and teardrop lysis	**Jumbo cup + porous metal augments (trabecular metal)** or structural allograft
IIIB ("up and in")	Over 60% rim loss	**Non-supportive**	**Kohler line breached**, severe ischial lysis, teardrop obliterated; **pelvic discontinuity** risk	**Cup-cage**, **custom triflange**, **augments + cage**, **distraction technique**

- **Pelvic discontinuity:** separation of the superior and inferior hemipelvis through the acetabulum (**fracture line through both columns, medial translation/rotation of the inferior hemipelvis, broken Kohler line**). Treat with **posterior column plating + cup**, **cup-cage**, **acetabular distraction**, or **custom triflange**.

- **AAOS classification:** I segmental, II cavitary, III combined, IV pelvic discontinuity, V arthrodesis.

Femur: Paprosky

Type	Bone loss	Reconstruction
I	Minimal metaphyseal loss (like primary)	Primary-type stem (cemented or uncemented)
II	**Metaphyseal loss**, **intact diaphysis**	**Extensively porous-coated (cylindrical) stem** or **modular tapered stem**
IIIA	Metaphysis non-supportive, **over 4 cm of intact isthmus** for scratch fit	**Extensively porous-coated stem** (diaphyseal fixation) or **modular tapered fluted (Wagner-type)**
IIIB	**Under 4 cm** of intact isthmus	**Modular tapered fluted titanium stem**
IV	**Extensive diaphyseal loss, widened canal**, no isthmus support	**Impaction allografting (cemented, Exeter technique)**, **allograft-prosthesis composite (young)**, **proximal femoral replacement (elderly)**

- **Extensively coated cylindrical cobalt-chromium stems** fail when the canal is **over about 18-19 mm** (stiff, stress shielding, less isthmus); **tapered fluted titanium stems** (modular or monoblock) are now widely used for most femoral revisions (reliable distal fixation, rotational stability via flutes).
- **Extended trochanteric osteotomy (ETO):** for **removing well-fixed stems/cement**, correcting proximal deformity (varus remodelling); length about **10-15 cm** (two-thirds of the trochanter with the lateral cortex); fix with cables; allows distal fixation **at least 4-6 cm below the osteotomy**.

Knee: Anderson Orthopaedic Research Institute (AORI)

Type	Bone loss	Reconstruction
1	**Intact metaphysis**, small defects	**Cement ± screws**, standard components
2A	Damaged metaphysis, **one condyle/plateau**	**Augments (metal wedges/blocks)**, particulate graft

Type	Bone loss	Reconstruction
2B	Damaged metaphysis, **both condyles/plateaus**	Augments, **cones/sleeves**
3	**Deficient metaphysis**, often with collateral ligament/tendon detachment	**Metaphyseal cones (porous tantalum/titanium) or sleeves**, **structural allograft**, **hinged prosthesis**, **distal femoral/proximal tibial replacement**

- **Three-zone fixation (Morgan-Jones): epiphysis (zone 1), metaphysis (zone 2), diaphysis (zone 3)**: aim for **solid fixation in at least two zones**; **metaphyseal fixation (cones/sleeves)** is key.
- **Stems: cemented** (shorter, fills canal) or **press-fit diaphyseal-engaging** (offset couplers to avoid tibial/femoral conflict).

4. Revision TKA principles

1. **Exposure:** medial parapatellar arthrotomy, synovectomy, release scar in the gutters, **external rotation of the tibia and medial release**; for difficult exposure: **quadriceps snip** (first choice; proximal oblique incision of the quadriceps tendon), **V-Y quadricepsplasty (turndown)** (extensor lag), **tibial tubercle osteotomy (TTO)** (long osteotomy, about 6-8 cm, fixed with wires/screws). **Avoid patellar tendon avulsion** (catastrophic): pin the tendon, use the snip/TTO early.
2. **Component removal:** preserve bone (osteotomes, saws, ultrasonic tools).
3. **Restore the tibial platform** first (establishes the joint line), then the **femur in flexion (size, rotation, posterior augments)**, then **extension**.
4. **Joint line restoration:** about **2.5 cm (one fingerbreadth) above the fibular head**, about **3 cm distal to the medial epicondyle**, **1 cm below the inferior pole of the patella**; avoid elevation (patella baja, mid-flexion instability).
5. **Constraint as needed** (PS, VVC, rotating hinge) to achieve stability with balanced gaps.
6. **Patella:** revise if loose/worn; patellar bone loss: **gull-wing osteotomy**, porous metal patellar augments, patelloplasty.

5. Revision THA principles

- **Approach:** usually posterior/extensile; **ETO** for stem/cement removal.
- **Acetabulum: uncemented hemispherical cups with screws** handle most defects (Paprosky I-IIIA), with **porous metal augments** for segmental loss; **impaction grafting with cemented cup** (Slooff) for contained defects; **cages/cup-cage/triflange** for IIIB/discontinuity.
- **Femur:** as per Paprosky; **cement-in-cement revision** (well-fixed intact cement mantle: re-cement a new taper-slip stem into the old mantle) for rapid revision with retained cement.
- **Instability prevention in revision: larger heads, dual-mobility cups** (reduce dislocation markedly in revision), **constrained liners** for abductor deficiency (high failure), **abductor reconstruction**.
- **Periprosthetic fracture revision** (Vancouver B2/B3) as in THA notes.

6. Complications of revision

- **Infection (higher than primary)**, **dislocation (up to about 10-15% in revision THA)**, **intraoperative fracture/perforation**, nerve injury, **aseptic re-loosening**, **extensor mechanism failure (TKA)**, wound breakdown (use previous lateral incision; **plastic surgery** input; medial gastrocnemius flap for exposed TKA), **thromboembolism**, **blood loss** (tranexamic acid, cell saver), **medical complications** (elderly).

7. Megaprostheses and salvage

- **Proximal femoral replacement (PFR)** / **distal femoral replacement (DFR)** / **total femur**: tumour resections, massive bone loss, elderly periprosthetic fractures (allows immediate weight bearing). **High dislocation (PFR)** and **infection rates (about 10%+)**; use **silver-coated** implants, mesh for soft-tissue attachment.
- **Allograft-prosthesis composite (APC):** younger patients (biological soft-tissue attachment).
- **Knee arthrodesis** (failed multiple revisions/infection with extensor loss), **above-knee amputation**, **resection arthroplasty** (Girdlestone hip).

Exam pearl: **Every failing arthroplasty is infected until proved otherwise. TKA: infection = commonest early failure; THA long-term: aseptic loosening/osteolysis.**

Exam pearl: **Paprosky acetabulum: II (rim/columns supportive, under 3 cm migration) = hemispherical cup; IIIA (up and out, Kohler intact) = jumbo cup + augments; IIIB (up and in, Kohler broken) = cup-cage/triflange; watch for pelvic discontinuity.**

Exam pearl: **Paprosky femur: II/IIIA = extensively porous-coated or modular tapered; IIIB (under 4 cm isthmus) = modular tapered fluted; IV = impaction grafting/APC/PFR. AORI knee: 1 cement, 2 augments/cones, 3 cones/sleeves/hinge/replacement; fix in two of three zones. Revision TKA exposure: quadriceps snip first; protect the patellar tendon.**

Shoulder arthroplasty

1. Types of shoulder arthroplasty

Type	What is replaced	Key requirement	Main indications
Hemiarthroplasty	Humeral head only	Functioning cuff; glenoid acceptable	**Young patients** with AVN (no glenoid involvement), unreconstructable proximal humerus fractures in younger patients (now largely replaced by reverse in elderly), glenoid bone too poor for a component
Resurfacing (humeral head)	Humeral head surface	Adequate humeral bone stock	Young, active patients; AVN; preserves bone
Anatomic total shoulder arthroplasty (TSA)	Head + **polyethylene glenoid**	**Intact, functioning rotator cuff**	**Primary glenohumeral osteoarthritis** (best results), inflammatory arthritis with intact cuff
Reverse total shoulder arthroplasty (RTSA)	**Glenosphere on glenoid, concave humeral cup**	**Functioning deltoid** and axillary nerve	**Cuff tear arthropathy**, massive irreparable cuff tears with pseudoparalysis, **proximal humerus fractures in the elderly (over about 70)**, failed arthroplasty, tumour, **B2/B3/C glenoids** with severe posterior erosion

2. Primary glenohumeral osteoarthritis

- **Pattern: posterior glenoid wear**, **posterior humeral head subluxation**, **anterior capsule and subscapularis contracture** (loss of **external rotation**: the commonest motion lost), **inferior humeral osteophyte ("goat's beard")**.

- **Walch classification of glenoid morphology (on axial CT):**
  - **A1** central minor erosion; **A2** central major erosion (medialisation),
  - **B1** posterior joint space narrowing, subchondral sclerosis, osteophytes; **B2 biconcave glenoid** (posterior erosion, posterior subluxation); **B3** monoconcave, posterior bone loss, **retroversion over about 15 degrees** and/or subluxation over 70%,
  - **C dysplastic glenoid, retroversion over 25 degrees**, not due to erosion,
  - **D** anteversion or anterior subluxation.
- **TSA outperforms hemiarthroplasty** for OA (pain relief, function, lower revision for glenoid erosion/pain). Correct moderate retroversion with **eccentric (high-side) reaming** (up to about 10-15 degrees), **augmented glenoid components**, bone grafting; severe B2/B3: consider **RTSA** (especially elderly).
- **Glenoid component: cemented all-polyethylene, pegged** components have lower radiolucency than keeled ones. **Glenoid loosening** is the **commonest long-term failure** of anatomic TSA (**rocking-horse phenomenon**: eccentric loading of the glenoid by a superiorly migrating head with cuff dysfunction).

3. Reverse total shoulder arthroplasty

Biomechanics (Grammont principles)

- **Medialises the centre of rotation** (onto the glenoid surface): increases the **deltoid moment arm** (more deltoid fibres recruited for abduction/elevation) and reduces shear on the glenoid component.
- **Lowers (distalises) the humerus**: **tensions the deltoid** (better lever).
- **Semi-constrained, fixed fulcrum**: allows elevation **without a functioning rotator cuff**.
- **Downsides of the classic Grammont design (155 degree neck-shaft angle, medialised): scapular notching**, limited rotation, loss of shoulder contour. **Modern designs lateralise** (glenoid-side via bony increased-offset RSA/BIO-RSA or metal augments; or humeral-side with a **135-145 degree** neck-shaft angle) to reduce notching, improve rotation and cuff (teres minor) tension.

Requirements

- **Functioning deltoid** and **intact axillary nerve** (the single absolute requirement), adequate glenoid bone stock.
- **External rotation:** depends on the **teres minor** (and infraspinatus). **Absent teres minor (positive hornblower's sign, external rotation lag)**: consider **latissimus dorsi (± teres major) transfer** with RTSA (L'Episcopo-type).

Complications of RTSA

- **Scapular notching** (commonest radiological finding; **Sirveaux classification grades 1-4**: erosion of the inferior scapular neck by the humeral cup in adduction; reduced by **inferior glenosphere placement, inferior tilt, lateralisation, larger glenosphere, lower neck-shaft angle**).
- **Instability/dislocation** (commonest complication in some series; **subscapularis deficiency**, inadequate soft-tissue tension, **medialised designs**, deltopectoral approach).
- **Acromial and scapular spine stress fractures** (**Levy classification**: type 1 acromion, type 2 at the base of the acromion/scapular spine origin of the deltoid, type 3 scapular spine): sudden pain and loss of elevation after a period of good function; associated with **over-lengthening (over-tensioning the deltoid), lateralisation**, osteoporosis.
- **Infection** (higher than anatomic TSA): ***Cutibacterium (Propionibacterium) acnes*** (**commonest in shoulder PJI**; **male sex**, young, hairy skin, sebaceous glands; slow-growing: **hold cultures for 14 days**; **benzoyl peroxide** skin preparation reduces load).
- **Glenoid baseplate loosening**, **nerve injury** (**axillary nerve**; brachial plexus traction with lengthening), **haematoma**, **humeral fracture**.

4. Proximal humerus fractures and arthroplasty

- **Hemiarthroplasty** results depend on **tuberosity healing** in the correct position (tuberosity malunion/non-union causes poor elevation): technically demanding; unpredictable function.
- **RTSA** gives **more reliable forward elevation and lower revision** than hemiarthroplasty in **elderly patients (over about 70) with 3/4-part fractures, head-split, fracture-dislocations**; **repair tuberosities** anyway

(improves external rotation and active elevation). **PROFHER trial:** no advantage of surgery overall over non-operative management for most displaced proximal humerus fractures (in elderly with surgical neck involvement).

- **Hertel (LEGO) criteria** for **head ischaemia** after fracture: **metaphyseal head extension under 8 mm**, **disrupted medial hinge (over 2 mm displacement)**, **anatomical neck fracture pattern**: high predictive value when combined.

5. Cuff tear arthropathy (CTA)

- Massive chronic cuff tear: superior migration of the humeral head, **acetabularisation of the acromion** (concave undersurface), **femoralisation of the humeral head** (rounding of the tuberosities), loss of the acromiohumeral interval (**under 7 mm**: suggests massive cuff tear), glenohumeral arthritis, **Milwaukee shoulder** (hydroxyapatite crystal-associated, elderly women, large blood-stained effusions).
- **Hamada classification: grade 1** AHI over 6 mm; **2** AHI 5 mm or less; **3** acetabularisation; **4** glenohumeral arthritis (4A without, 4B with acetabularisation); **5** humeral head collapse.
- **Seebauer classification** (biomechanics: centred vs decentred, stable vs unstable): **IA, IB** (centred), **IIA, IIB** (decentred: superior migration / anterosuperior escape: IIB = **anterosuperior escape** with **coracoacromial arch** loss: **avoid acromioplasty/CA ligament release** in massive tears!).
- **RTSA** is the treatment of choice for symptomatic CTA with pseudoparalysis; **hemiarthroplasty with a CTA head** (extended articular surface) in younger/low-demand patients with preserved elevation and intact coracoacromial arch.

6. Other indications and specifics

- **Rheumatoid arthritis:** central glenoid erosion (medialisation), cuff tears common; **RTSA** if cuff deficient; check the cervical spine.
- **AVN of the humeral head (Cruess classification, stages I-V, like Ficat):** steroids, sickle cell, alcohol, post-trauma; **hemiarthroplasty/resurfacing** if the glenoid is intact (stage III-IV), **TSA** if glenoid involved (stage V).

- **Instability arthropathy** (after previous stabilisation, e.g. **capsulorrhaphy arthropathy** from over-tight anterior repair (Putti-Platt), **hardware arthropathy**).
- **Post-infective arthritis**, **post-traumatic arthritis** (fracture sequelae: **Boileau classification** types 1-4; **types 3 (non-union of the surgical neck) and 4 (severe tuberosity malunion)** do poorly with anatomic arthroplasty unless osteotomy: favour **RTSA**).
- **Approach: deltopectoral** (commonest; cephalic vein, **subscapularis management: tenotomy, peel or lesser tuberosity osteotomy**; subscapularis failure after anatomic TSA: anterior instability, weakness: **lift-off and belly-press signs**). **Superolateral (Neviaser-Mackenzie)** approach for some RTSA (less instability, worse glenoid exposure, risk to the axillary nerve with deltoid split over about 5 cm distal to the acromion).
- **Stemless** humeral components (metaphyseal fixation) in anatomic TSA: preserve bone.
- **Rehabilitation:** protect subscapularis repair (limit external rotation initially) in anatomic TSA.

7. Outcomes

- **Anatomic TSA for OA:** about **90%+ survival at 10 years**; best outcomes of any shoulder arthroplasty for pain relief.
- **RTSA:** good pain relief and elevation for CTA (**forward elevation typically to about 120-140 degrees**), **limited internal and external rotation**; survival about 90% at 10 years; worse outcomes in **revision RTSA** and post-traumatic sequelae.
- **Hemiarthroplasty for OA: progressive glenoid erosion** (painful), inferior to TSA.

> Exam pearl: **Anatomic TSA needs an intact cuff; RTSA needs a working deltoid (axillary nerve). Commonest late failure of anatomic TSA = glenoid loosening (rocking horse). Commonest radiological finding after RTSA = scapular notching.**

Exam pearl: **Grammont: medialise and distalise the centre of rotation to increase the deltoid lever arm. Loss of teres minor (hornblower's sign): add latissimus dorsi transfer. Acromial stress fracture after RTSA: sudden pain and loss of elevation.**

Exam pearl: **Shoulder PJI: C. acnes; hold cultures 14 days. Walch B2 = biconcave glenoid. Elderly 3/4-part proximal humerus fracture: RTSA more reliable than hemiarthroplasty (tuberosity healing).**

Unicompartmental and osteotomy

1. The spectrum for knee osteoarthritis

- **Non-operative first:** weight loss, quadriceps strengthening and exercise (strong evidence), analgesia (paracetamol, topical/oral NSAIDs), intra-articular steroid (short-term), bracing (unloader), walking aid in the contralateral hand.
- **Arthroscopic debridement/lavage** for degenerative knees: **no benefit** (Moseley 2002, Kirkley 2008); **arthroscopic partial meniscectomy for degenerative tears** is no better than physiotherapy or sham (**FIDELITY**).
- **Joint-preserving surgery** (osteotomy) for **young, active patients with unicompartmental disease and malalignment**; **unicompartmental knee arthroplasty (UKA)** for isolated compartment disease in suitable patients; **TKA** for tricompartmental or advanced disease.

2. High tibial osteotomy (HTO)

Rationale

- **Varus knee + medial compartment OA**: shift the **mechanical axis (weight-bearing line)** from the diseased medial compartment to the **lateral compartment**: the target is the **Fujisawa point**, about **62-66% of the tibial plateau width** from medial (**30-40% beyond the midpoint, i.e. slight overcorrection into about 3-5 degrees of mechanical valgus**).

Patient selection (ideal)

- **Age under about 55-60**, active/heavy labour, **BMI under about 30**.
- **Isolated medial compartment OA** (Ahlback grades I-II; Kellgren-Lawrence under 4), **intact lateral compartment and patellofemoral joint** (mild PFJ changes acceptable).
- **Range of motion: flexion over about 90-100 degrees**, **fixed flexion deformity under about 10-15 degrees**.
- **Varus deformity under about 10-15 degrees**, **metaphyseal varus (tibial origin: medial proximal tibial angle (MPTA) under 85 degrees)**: the deformity must be in the tibia to correct in the tibia.

- **Stable knee** (ACL deficiency can be addressed with slope change: reduce slope for ACL-deficient knees; increase slope for PCL-deficient knees).
- **Contraindications:** inflammatory arthritis, **lateral compartment disease or meniscectomy**, **tibial subluxation over 1 cm**, severe PFJ OA, poor ROM, smokers (non-union).

Planning

- **Long-leg standing radiographs (hip-to-ankle):** mechanical axis, **Miniaci method** (draw the desired weight-bearing line through the Fujisawa point, rotate around the hinge point to calculate the correction angle/wedge), **joint line convergence angle (JLCA, soft-tissue laxity: overcorrection risk if JLCA over 2-3 degrees)**. Ensure post-op **MPTA under about 95 degrees** (avoid excessive joint line obliquity: shear on cartilage) and decide whether a **double-level osteotomy** (femur + tibia) is needed for combined deformity.

Techniques

	Medial opening-wedge HTO	**Lateral closing-wedge HTO (Coventry)**
Approach	Medial; osteotomy from medial towards the **lateral hinge** (about 1 cm from the lateral cortex, towards the tip of the fibular head)	Lateral; wedge removed; **fibular osteotomy or proximal tibiofibular joint disruption** needed
Advantages	**No fibular osteotomy**, **no peroneal nerve risk**, adjustable correction intraoperatively, preserves bone stock (easier later TKA), can address the **sagittal plane**	Bone-on-bone contact: faster union, early weight bearing, no graft

	Medial opening-wedge HTO	Lateral closing-wedge HTO (Coventry)
Disadvantages	**Increases posterior tibial slope** (if the gap is not made trapezoidal: gap posteromedially about twice anteromedially), **patella baja** (biplanar osteotomy behind the tubercle avoids it), **graft/bone substitute** for larger corrections, **delayed union**, **lateral hinge fracture** (Takeuchi types I-III: type II/III unstable: loss of correction)	**Peroneal nerve injury**, **leg shortening**, **decreases slope**, **patella baja (less)**, **lateral tibial bone loss** complicates later TKA (lateral offset, difficult exposure)
Fixation	**Angle-stable locking plate (TomoFix)**	Staples/plate

- **Outcomes:** about **75-90% survival at 10 years** (conversion to TKA as endpoint); worse with older age, higher BMI, advanced OA, undercorrection (**undercorrection is the commonest cause of failure**).
- **Complications:** under/overcorrection, loss of correction, non-union (opening-wedge, smokers), **lateral hinge fracture**, **peroneal nerve palsy (closing wedge)**, **popliteal artery injury** (protect posteriorly with a retractor, keep the knee flexed), infection, compartment syndrome, DVT, **change in posterior tibial slope**, patella baja.

3. Distal femoral osteotomy (DFO)

- **Valgus knee + lateral compartment OA**, **femoral-origin valgus (lateral distal femoral angle (LDFA) under about 85-87 degrees)**: **varus-producing DFO**.
- **Medial closing-wedge** (stable, faster union) or **lateral opening-wedge** (easier, graft needed; **IT band irritation** from plate). Target: mechanical axis to about **50% (neutral)** or slightly medial (**do not overcorrect into varus**: unlike HTO).
- Also used to unload **lateral osteochondral grafts/meniscal transplant**, correct **patellofemoral malalignment**.
- **Valgus knee correction in the tibia** is avoided because it creates **excessive joint line obliquity** (valgus is usually femoral).

4. Other osteotomies around the knee (brief)

- **Tibial tubercle osteotomy (Fulkerson anteromedialisation)** for patellofemoral malalignment/lateral facet arthritis (see sports notes).
- **Slope-changing (anterior closing-wedge) osteotomy** for revision ACL with **posterior tibial slope over about 12 degrees**.
- **Double-level osteotomy** for severe deformity in both bones (to keep the joint line horizontal).
- **Opening-wedge HTO + ACL reconstruction** for varus ACL-deficient knees (**double/triple varus**: Noyes classification of varus: primary (bony varus), double (+ lateral tibiofemoral separation/LCL laxity), triple (+ posterolateral corner insufficiency, varus recurvatum)).

5. Unicompartmental knee arthroplasty (UKA)

Classical indications (Kozinn and Scott, 1989) vs the Oxford criteria

- **Kozinn and Scott:** age **over 60**, **weight under 82 kg**, low demand, **flexion over 90 degrees**, **fixed flexion under 5 degrees**, **angular deformity under 15 degrees (correctable passively)**, **no inflammatory arthritis**, **no chondrocalcinosis**, intact ACL, **no PFJ exposed bone**.
- **Oxford group (Goodfellow, Murray) broadened: anteromedial OA** (the classic disease pattern: **bone-on-bone medial with full-thickness cartilage preserved laterally and posteriorly in the medial compartment**, **functionally intact ACL**, **correctable varus**, **intact MCL**), **regardless of age, weight, activity, chondrocalcinosis and PFJ status (unless lateral PFJ bone loss with grooving/subluxation)**. **Spontaneous osteonecrosis (SONK/SIFK)** of the medial condyle is also an indication.
- **Contraindications:** inflammatory arthritis, **ACL deficiency** (for mobile-bearing: fixed-bearing UKA ± ACL reconstruction in young patients), **lateral compartment disease** (for medial UKA), **fixed deformity not correctable**, previous HTO (relative), **lateral PFJ damage with bone loss**.
- **Pre-op workup: valgus stress radiograph** (lateral compartment full-thickness cartilage preserved: lateral joint space maintained) and **varus stress radiograph** (bone-on-bone medially: confirms full-thickness loss; **avoid UKA for partial-thickness disease**: high rate of unexplained pain/revision).

Designs

	Mobile-bearing (Oxford)	Fixed-bearing (metal-backed or all-polyethylene)
Conformity	**Fully congruent spherical femur and bearing**: low contact stress, **minimal polyethylene wear**	Less conformity: higher contact stress
Main issue	**Bearing dislocation** (especially **lateral UKA: about 10%** mobile-bearing; use fixed-bearing laterally or a domed lateral design)	Polyethylene wear, loosening
Requirements	**Intact, functioning ligaments** (ACL, MCL)	Slightly more forgiving of ligaments

- **Robotic-assisted UKA** improves component positioning accuracy.

Principles of surgery

- **Do not overcorrect** the deformity: **correct to the pre-disease alignment**, not neutral (**overcorrection overloads the opposite compartment: progression of OA there**); leave the **MCL unreleased** (no soft-tissue release in medial UKA).
- **Minimally invasive approach**; preserve the ACL.

Outcomes and failures

- **Advantages over TKA:** faster recovery, **more natural kinematics and better ROM**, lower morbidity (**lower mortality, VTE, MI, infection, transfusion** in registry analyses), better function ("forgotten knee"), less blood loss, shorter stay.
- **Disadvantages: higher revision rate in registries** (about **2-3 times that of TKA**), partly due to the low threshold for revision (a UKA is easier to revise than a TKA), surgeon volume (**surgeons doing over about 20% of their knee arthroplasties as UKA**, or 13+ per year, have better results); **TOPKAT trial** (UKA vs TKA in medial OA): **similar outcomes and revision at 5 years**, UKA more cost-effective.

- **Causes of failure: progression of OA in the other compartments** (lateral), **aseptic loosening (tibial)**, **unexplained pain**, **bearing dislocation** (mobile), polyethylene wear (fixed), **tibial plateau fracture** (usually early: excessive vertical saw cut, pin holes), infection.
- **Radiolucent lines** under the tibial component of the Oxford UKA (**physiological, thin, well-defined, under 1-2 mm, non-progressive**) are **common and do not indicate loosening**: **do not revise** for them alone.
- **Lateral UKA:** less common (**about 5-10% of UKAs**); lateral compartment has more **rollback and laxity** in flexion; valgus deformity often has a femoral component.

6. Patellofemoral arthroplasty (PFA)

- **Isolated patellofemoral OA** (often **trochlear dysplasia**-related) in younger patients; **onlay** trochlear designs have fewer patellar maltracking problems than **inlay** designs.
- **Failure: progression of tibiofemoral OA** (commonest), maltracking, pain; convert to TKA.

7. Bicompartmental and other options

- **Bi-unicompartmental** and **bicompartmental (medial UKA + PFA)** arthroplasty: niche.
- **Cartilage procedures** for focal defects (see sports notes) are distinct from OA.

8. HTO vs UKA in medial OA (choosing)

Feature	Favours HTO	Favours UKA
Age	Younger (**under 55-60**)	Older (**over 60**) (Oxford: any age)
Activity	**High (manual labour, sports)**	Lower to moderate
Deformity	**Tibial (metaphyseal) varus, constitutional**	Intra-articular varus from cartilage loss (correctable)
OA severity	**Mild-moderate (partial thickness)**	**Bone-on-bone (full-thickness)**, SONK

Feature	Favours HTO	Favours UKA
ACL	Can be deficient (combine with ACL reconstruction)	**Must be intact** (mobile-bearing)
ROM	Good (flexion over 90)	Good

Exam pearl: **HTO: young, active, isolated medial OA, tibial varus (MPTA under 85); target the Fujisawa point (62-66% of plateau width). Opening-wedge increases slope and causes patella baja; closing-wedge risks the peroneal nerve. Commonest cause of HTO failure: undercorrection.**

Exam pearl: **DFO for valgus knee with femoral deformity: correct to neutral, not overcorrected. Valgus is corrected on the femoral side to avoid joint line obliquity.**

Exam pearl: **Oxford UKA: anteromedial OA with full-thickness medial loss, functioning ACL, correctable varus, intact lateral compartment (valgus stress view). Do not overcorrect; do not release the MCL. Physiological tibial radiolucent lines are not loosening. Mobile-bearing lateral UKA = bearing dislocation. UKA: higher registry revision but lower morbidity than TKA (TOPKAT similar).**